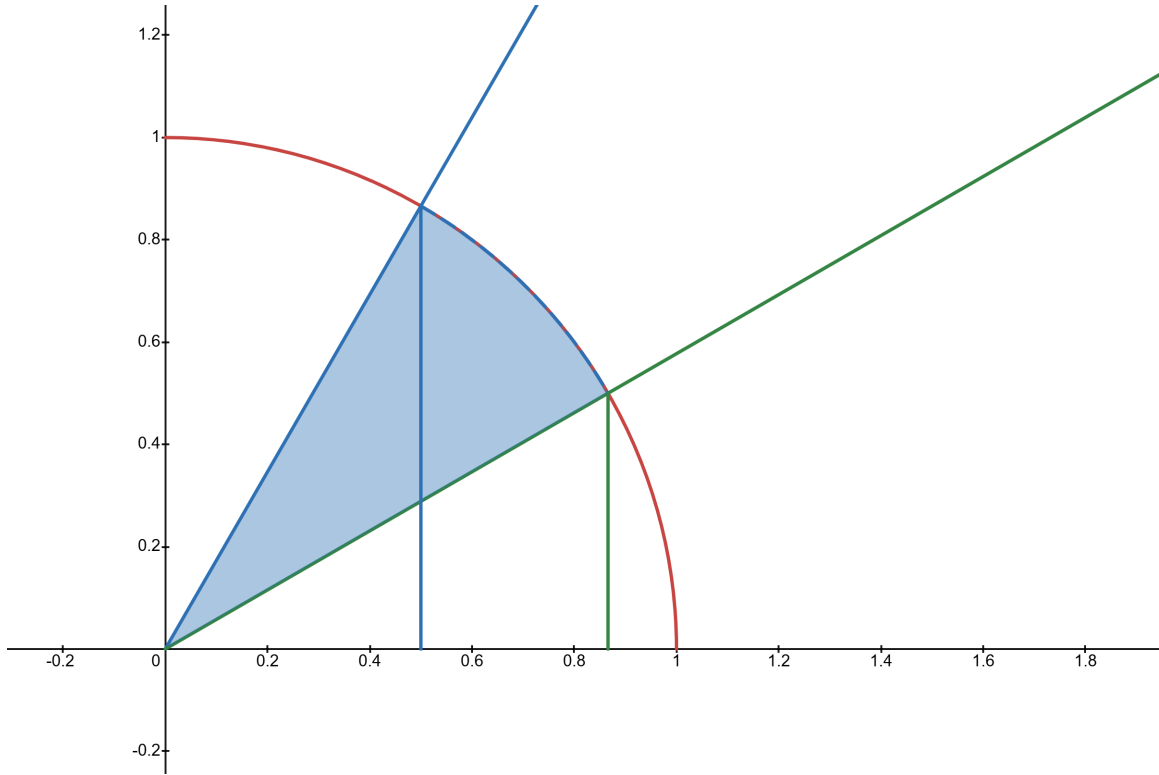


Another Bertrand Paradox Problem

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Problem 1. What is the probability that a perimeter point of quarter-circumference is in the highlighted part?



- $x^2 + y^2 = 1$
- $x = \tan(\frac{\pi}{6}) \cdot y$
- $x = \tan(\frac{\pi}{3}) \cdot y$

Proof 1. Let us think about the angle θ . The affected arc of circumference is reached when $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$

$$p_1 = \mathbb{P}(\{\theta \in (\frac{\pi}{6}, \frac{\pi}{3})\}) = \frac{\frac{\pi}{3} - \frac{\pi}{6}}{\frac{\pi}{2} - 0} = \frac{1}{3}$$

Proof 2. Let us think about x or y . The affected arc of circumference is reached when $\cos(\frac{\pi}{3}) < x < \cos(\frac{\pi}{6})$ and $y \geq 0 \text{ ' } x^2 + y^2 = 1$.

$$p_2 = \mathbb{P}(\{\cos(\frac{\pi}{3}) < x < \cos(\frac{\pi}{6})\}) = \frac{\cos(\frac{\pi}{6}) - \cos(\frac{\pi}{3})}{1 - 0} = \frac{\frac{\sqrt{3}}{2} - \frac{1}{2}}{1} = \frac{\sqrt{3} - 1}{2} \approx 0.366$$

It is easily observable that $p_2 > p_1 \Rightarrow p_2 \neq p_1$

$$\Delta\mathbb{P} = |p_2 - p_1| \approx 0.03269 \approx 3.27\%$$